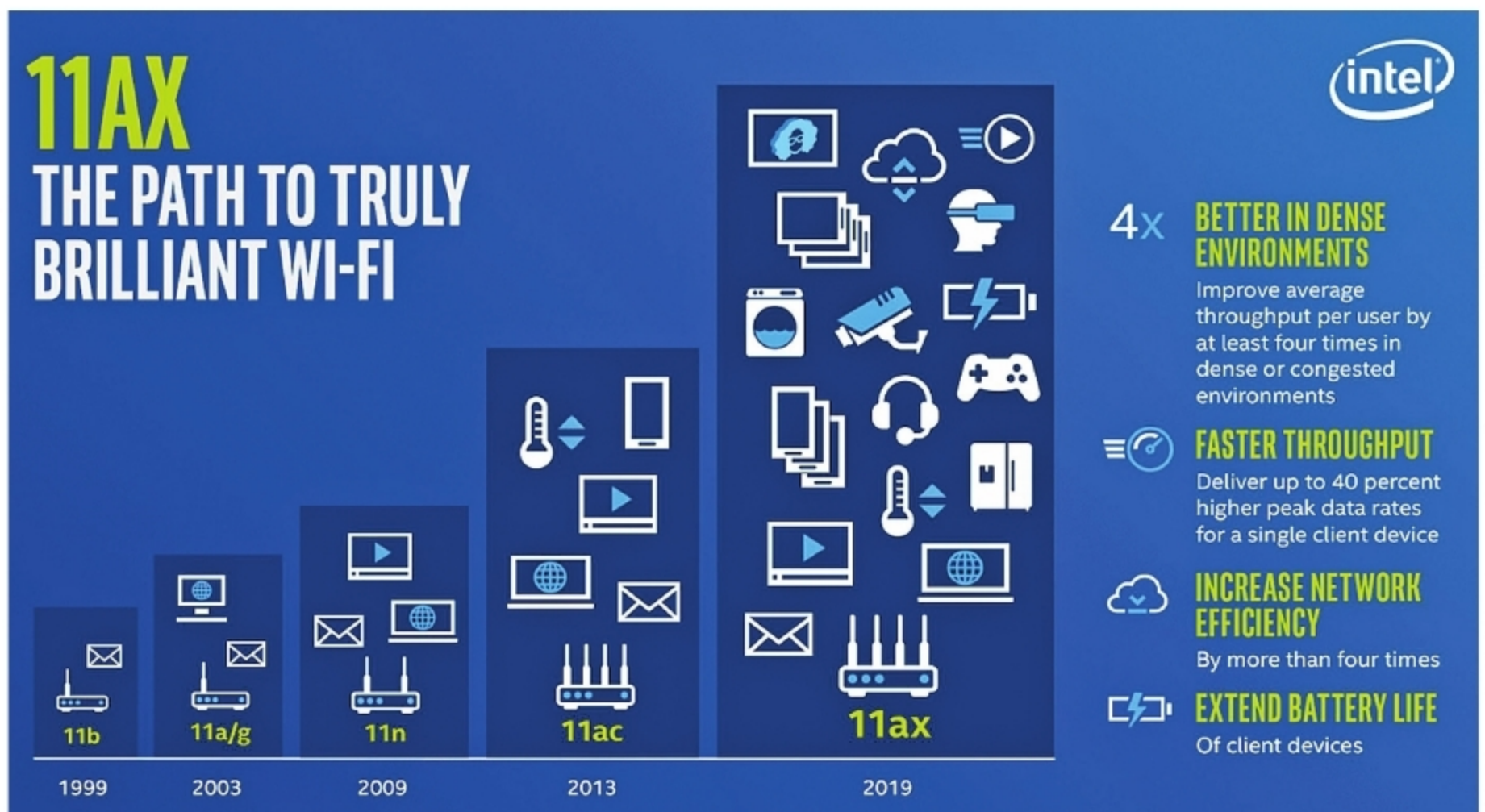


Wireless Communication

Keeping Pace with Next-Gen Technologies

To keep pace with new technologies like the Internet of Things, artificial intelligence and 5G spectrum, wireless technologies are also getting upgraded. Here is a peek at what's happening in the areas of short-range as well as long-range communication.



Features of Wi-Fi 6 (Credit: www.howtogeek.com)

For a connected world we need various short- and long-range wireless communication devices to solve challenges like pollution control, disaster management, infrastructure monitoring, resource management and waste management. Short-range wireless communication can transfer data from a few centimetres to several metres. Some technologies that support short-range wireless communication

are Bluetooth, Wi-Fi, near-field communications (NFC) and Zigbee.

Long-range wireless communication uses signals that travel from a few to several thousand kilometres. Some technologies that support long-range wireless communication are radio transmission, satellite communication and LoRaWAN.

Bluetooth 5

Bluetooth replaces cables between

different electronic devices for their communication with each other by establishing a personal area network. It can transmit audio, video, images or documents at high speed within fifty metres range. In its setup, one device operates as the master and one or more devices operate as slaves to send and receive data.

Bluetooth 5 is the latest version of this technology. It consumes low power and transfers data at 2Mbps speed.